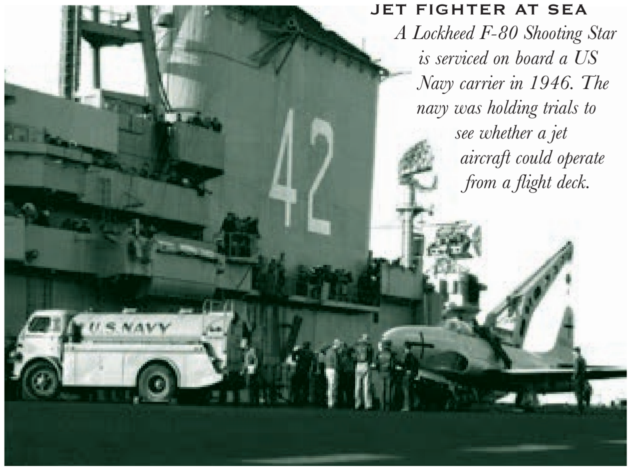
flight testing that attracted the young airmen with the best natural skills and the nerve to put their lives on the line. From the late 1940s through the 1950s, they would fly faster than any humans had ever traveled before, in aircraft that could never be predicted to perform safely. The wind tunnels and other forms of simulation available at the time were simply not good enough to allow designers to iron out the flaws in their aircraft or to model the aerodynamic problems of high-speed flight. Experiments had to be conducted for real, with a pilot at the controls.



JET FIGHTER AT SEA
A Lockheed F-80 Shooting Star is serviced on board a US Navy carrier in 1946. The navy was holding trials to see whether a jet aircraft could operate from a flight deck.

The key players

The development of military jets was a high-cost, high-technology business that was only going to be open to a very few players. After the Germans – clear leaders in jet-aviation technology – were put out of contention by defeat in the war, Britain found itself temporarily out in front in 1945, with its Gloster Meteor and de Havilland Vampire fighters. But the British did not have the resources to lead the way for long, and by the end of the decade they had been matched even in Europe, by the French with their Dassault Ouragan fighter and by Sweden's Saab 29 Tunnan. Inevitably, though, it was the United States and the Soviet Union that devoted the greatest resources to military jet development and that were soon reaping the rewards. The United States was surprisingly tardy in turning to jet propulsion. The Americans in effect let the Germans and British do the ground-breaking research and then built on the results.

The first US jet fighter, the Bell P-59 Airacomet, was constructed around British designer Frank Whittle's jet engine, and licensed to General Electric to build. It first flew in 1942, but performed disappointingly and was never sent into combat. The Lockheed P-80 Shooting Star, designed by Kelly Johnson's Skunk Works in 1943, also originally had a British engine, but only came into its own when refitted with the all-American Allison J33. As the F-80, the Shooting Star was the US's first operational jet fighter – although too late for service in World War II – and gave birth to the T-33 jet trainer, in which generations of American fighter pilots learnt the basics of their trade. The Soviet Union also used British engine technology in its jet-powered MiG-15s.

Airframe design

For crucial progress in airframe design, the Americans and Soviets learned from the Germans. The Me 262 had been designed with a partially swept-back wing. Captured documents showed that German aerodynamics experts had intended to sweep it back more fully, since their data indicated that this would reduce drag at very high speeds. They had not done so because a swept wing caused instability at low speeds, and they had no answer to this problem.

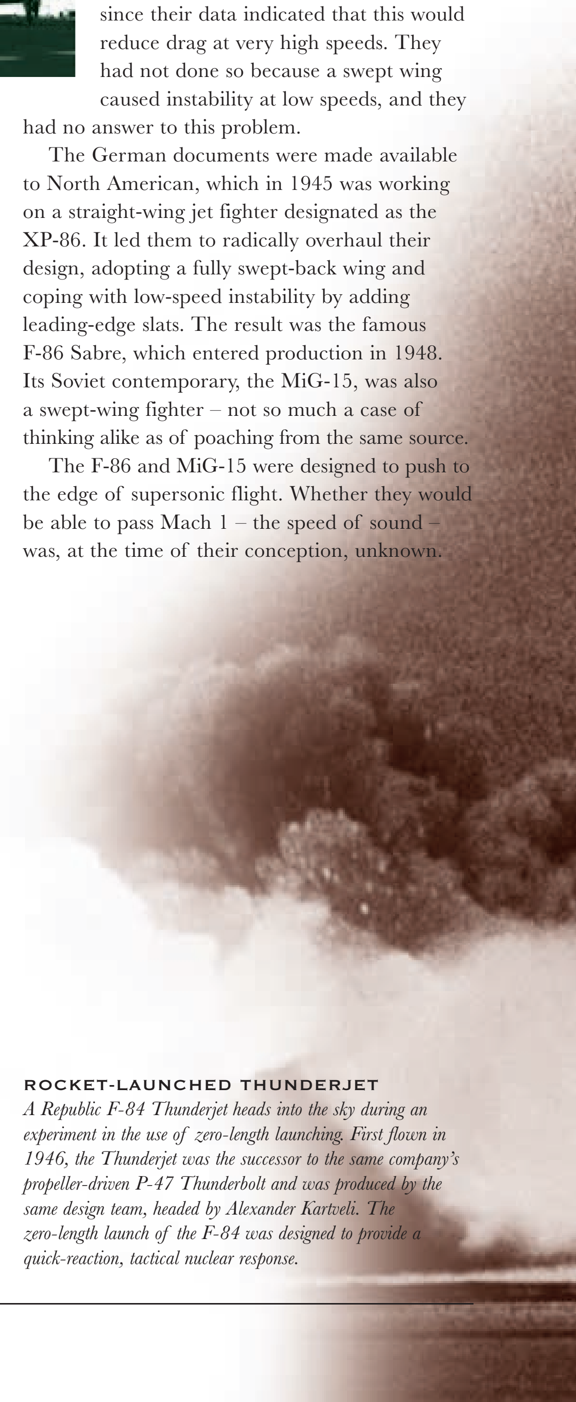
The German documents were made available to North American, which in 1945 was working on a straight-wing jet fighter designated as the XP-86. It led them to radically overhaul their design, adopting a fully swept-back wing and coping with low-speed instability by adding leading-edge slats. The result was the famous F-86 Sabre, which entered production in 1948. Its Soviet contemporary, the MiG-15, was also a swept-wing fighter – not so much a case of thinking alike as of poaching from the same source.

The F-86 and MiG-15 were designed to push to the edge of supersonic flight. Whether they would be able to pass Mach 1 – the speed of sound – was, at the time of their conception, unknown.

CLARENCE "KELLY" JOHNSON

INNOVATIVE DESIGNER
Kelly Johnson stands over a model of the Lockheed P-80 Shooting Star, the product of the original Skunk Works. Johnson's aircraft design work was characterized by a taste for radical innovation and imaginative solutions.

Aircraft designer Clarence "Kelly" Johnson (1910–90) joined the Lockheed company in 1933, armed with a degree in aeronautical engineering from the University of Michigan. He became Lockheed's chief engineer while still in his twenties, designing the P-38 fighter and the Constellation airliner. In 1943 Johnson led the project to build the P-80 jet fighter, taking it from initial design to first flight in seven months. He achieved this by setting up a small, tightly integrated team working in total secrecy with the minimum of bureaucratic interference. Because it had to put up with the smell from a nearby plastics factory, the team was dubbed the "Skunk Works." The Skunk Works concept – defined by Johnson as "a few good people solving problems far in advance… by applying the simplest and most straightforward methods possible" – was subsequently applied to a string of Johnson-designed Lockheed aircraft, including the F-104 Starfighter, and the U-2 and SR-71 high-altutude reconnaissance aircraft.



ROCKET-LAUNCHED THUNDERJET
A Republic F-84 Thunderjet heads into the sky during an experiment in the use of zero-length launching. First flown in 1946, the Thunderjet was the successor to the same company's propeller-driven P-47 Thunderbolt and was produced by the same design team, headed by Alexander Kartveli. The zero-length launch of the F-84 was designed to provide a quick-reaction, tactical nuclear response.